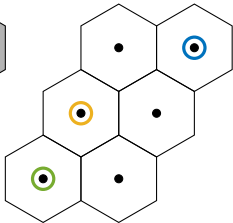
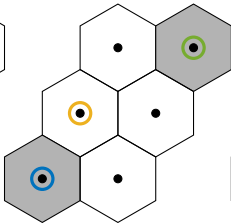
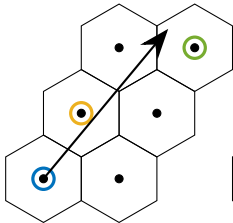


$$x_i - (\nabla^2 \Phi_i^a(x_i))^{-1} \nabla \Phi_i^a(x_i) + r$$

vertex i at x_i^{new}



vertex i at x_i

swapped vertex